4.5.1 (a) Check $\sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} E_k^{(0,d)} \in \mathcal{M}_k(\Gamma_0(N))$.

Pf: Let $f(\tau) = \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} E_k^{(0,d)}(\tau)$.

Then $f(\gamma\tau) = \sum_d E_k^{(0,ad)}(\tau)(c\tau+d)^k \quad \gamma\in\Gamma_0(N)$

$= (c\tau+d)^k \sum_{d'} E_k^{(0,d')}(\tau) = j(\gamma,\tau)^k f(\tau)$.

(b) Check $\sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \chi(d) E_k^{(0,d)}(\tau) \in \mathcal{M}_k(N,\chi)$.

Pf: Let $f_\chi(\tau) = \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \chi(d) E_k^{(0,d)}(\tau)$.

Then $f_\chi(\gamma\tau) = \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \chi(d) E_k^{(0,ad)}(\tau)(c\tau+d)^k \quad \gamma\in\Gamma_0(N)$.

$= \sum_{d'} \chi(\bar{a})\chi(d') E_k^{(0,d')}(\tau)(c\tau+d)^k$

$\overset{ad-bcN=1}{=} \chi(d)(c\tau+d)^k f_\chi(\tau)$.

(c) Verify calculations showing $G_k^{\psi,\varphi} \in \mathcal{M}_k(N,\psi\varphi)$.

Pf: $\gamma = \begin{pmatrix} a & b \\ c & d\end{pmatrix} \in \Gamma_0(N) \Rightarrow \overline{(cv, d+ev)}\gamma = \overline{(c'v, d'+e'v)}$

$c', e' \in \{0,\dots,u-1\}$ $\quad c' \equiv c a_\gamma \ (u)$

$d' \in \{0,\dots,v-1\}$ $\quad d' \equiv d d_\gamma \ (v)$

$e' \equiv (e + c' b_\gamma) d_\gamma \ (u)$.

Also $\psi(c)\bar{\varphi}(d) = (\psi\varphi)(d_\gamma)\psi(c')\bar{\varphi}(d')$.

Thus $G_k^{\psi,\varphi}[\gamma]_k(\tau) = \sum_{c,d,e} \psi(c)\bar{\varphi}(d) E_k^{\overline{(cv,d+ev)}\gamma}(\gamma\tau)(c\tau+d)^{-k}$

$= (\psi\varphi)(d_\gamma)\cancel{j(\gamma,\tau)^k} \sum_{c',d',e'} \psi(c')\bar{\varphi}(d') E_k^{\overline{(c'v,d'+e'v)}}(\tau) = (\psi\varphi)(d_\gamma) G_k^{\psi,\varphi}$

9.5.2 (a) If k even, const term of $G_k^{\psi,\varphi}(\tau)$

is $(C_k\, g(\bar\varphi)/v^k)\, \psi(0)\, L(1-k,\varphi)$.

Pf: Const is $\displaystyle\sum_{c=0}^{u-1}\sum_{d=0}^{v-1}\sum_{e=0}^{u-1} \psi(c)\,\bar\varphi(d)\, g(\bar v)\, \zeta^{\overline{d+ev}}(k)$

$$= \psi(0)\sum_{d=0}^{v-1}\sum_{e=0}^{u-1}\bar\varphi(d)\,\zeta^{\overline{d+ev}}(k) = \psi(0)\sum_{d=0}^{v-1}\bar\varphi(d)\,\zeta^{\bar d}(k)$$

$$= \psi(0)\sum_{d=0}^{v-1}\bar\varphi(d)\sum_{\substack{m\equiv d(v)\\ m\neq 0}}\frac{1}{m^k} = \psi(0)\sum_{m\neq 0}\frac{\bar\varphi(m)}{m^k} = 2\psi(0)L(k,\bar\varphi)$$

$$\doteq 2\psi(0)\cdot\frac{\cancel{-i}\,\pi^{-(1-k)/2}}{\pi^{-(k+1)/2}}\,\frac{\Gamma(\frac{1-k}{2})}{\Gamma(\frac{k+1}{2})}\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\bar\varphi)$$

$$= \psi(0)\, C_k\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\varphi).$$

(b) If k odd, const term of $G_k^{\psi,\varphi}(\tau)$ is the same

Pf: $$2\psi(0)L(k,\bar\varphi) = 2\psi(0)\,\frac{-i\pi^{-(1-k)/2}}{\pi^{-(k+1)/2}}\,\frac{\Gamma(\frac{1-k}{2})}{\Gamma(\frac{k+1}{2})}\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\bar\varphi)$$

$$\overset{k+1=l}{=} 2\psi(0)\,\frac{-i\pi^{-(l-1)/2}}{\pi^{-l/2}}\,\frac{\Gamma(\frac{l-1}{2})}{\Gamma(\frac{l+1}{2})}\,\frac{g(\bar\varphi)}{v^{k}}\,L(1-k,\bar\varphi)$$

$$= \psi(0)\,\frac{(-i)^{\frac{(l-1)}{2}}}{\pi}\,C_l\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\bar\varphi)$$

$$= \psi(0)\,\frac{(-i)^{\frac{(l-1)}{2}}}{\pi}\,\frac{(-2\pi i)^{k+1}}{k!}\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\bar\varphi) = \psi(0)\,C_k\,\frac{g(\bar\varphi)}{v^k}\,L(1-k,\bar\varphi).$$